

NT48A08 NT48B20 NT48C32 MODBUS RTU Command

MODBUS command (function code, write 06/16, read 04/03)

Note :

1 MODBUS command must be HEX

2 Slave ID (device address) must be consistent with the DIP switches (A0-A5)



3 The Baudrate and parity should be consistent

4 The default baud rate is 9600 ,8 Data bits,None Parity,1 Stop Bit.

5 If you forget the baud rate, please short the RES jumper for 5 seconds to restore the factory settings.

Command overview:

Channels	SKU	Product ID	Channel assignment
16	NT48A08	2608	8Ch B3950-10K NTC acquisition board
20	NT48A20	2620	20Ch B3950-10K NTC acquisition board
32	NT48C32	2532	32Ch B3950-10K NTC acquisition board

Supported function codes:

Function Code	Modbus Address (PLC)	Register Address	Describe
04:	30001	0x0000-0x0001F (0-31)	Unit 0.1℃ 1-32 channel NTC temperature value
		0x0020-0x0003F (32-63)	Unit 0.01KΩ 1-32 channel NTC resistance value For example, the data obtained is 1000, which is equal to 10.00KΩ
03	40001		
		0x0080-0x00FF (128-255)	Read special function registers (baud rate 485 address, etc.)
06	40001		
		0x0080-0x00FF (128-255)	Write a single special function register (baud rate 485 address, etc.)
16(0x10)	40001		
		0x0080-0x00FF (128-255)	Write multiple special function registers (baud rate 485 address, etc.)

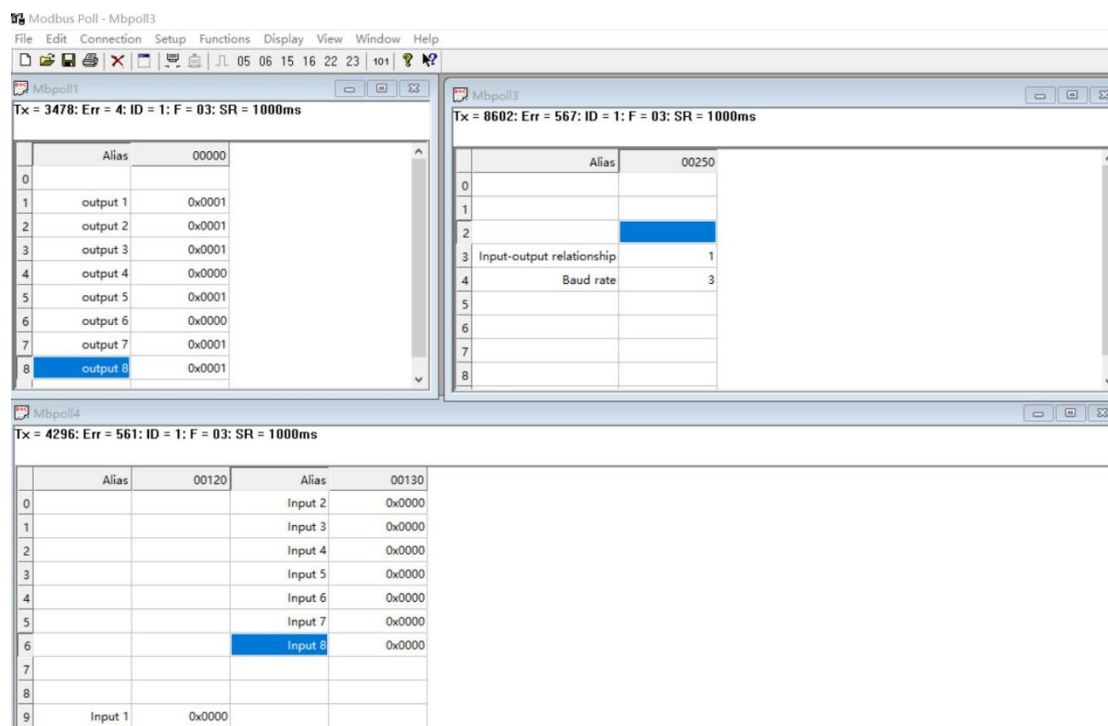
All states are mapped into 4xxxx range registers. The user can monitor the input and output status of the module by reading or modifying the value of the 4xxxx interval register (03 06 16 function code)

Register address	Register contents	Register value	Remarks	R/W
0x0000-0x001F (0-31)	Unit 0.1°C 1-32 channel NTC temperature value			R
0x0020-0x003F (32-63)	Unit 0.01KΩ 1-32 channel NTC resistance value For example, the data obtained is 1000, which is equal to 10.00K Ω			R
0x0040-0x005F (64-95)	Channel 1-32 temperature correction value, unit 0.1°C 2 bytes, input positive number to increase temperature; input negative number to decrease temperature			R/W
0x0060-0x007F (96-127)	Channel 1-32 resistance correction value, unit 0.01KΩ 2 bytes, input positive resistance increases; input negative resistance decreases			R/W
0X00F6 (246)	Re-power on the board without any command operation (Equivalent to a watchdog)	0-65535 The board will be powered on again when no command is received within the set time. 0: Do not power on again (default) 1: Power on again after 1 second without any command 2: Power on again after 2 seconds without any command 10: Power on again after 10 seconds without any command You can set it to a maximum of 65535 seconds		R/W
0x00F7 (247)	Product ID	0-65535	SKU ID NT48A08 2608 NT48B20 2620 NT48C32 2632	
0x00FB (251)	Factory Reset	Factory Reset: 1 1 Short the RES jumper for 5 seconds 2 Enter the following command at the current baud rate: FF 06 00 FB 00 00 ED E5		R/W
0x00FC (252)	Command Return Time	0-1000	Time interval for command return (unit:MS) Setting	R/W

			value: 0-1000	
0x00FD (253)	RS485 address (Station address)	Read address: FF 03 00 FD 00 01 00 24; Set the RS485 address via the DIP switch on the board		R/W
0x00FE (254)	Baud rate	0-255	0:1200 1:2400 2:4800 3:9600 (default) 4:19200 5:38400 6: 57600 7: 115200 Others: Factory reset	R/W
0x00FF (255)	Parity	0-2	0 None Parity 1 Even Parity 2 Odd Parity	R/W

9600 Band ,8 Data bits,None Parity,1 Stop Bit。

MODBUS commands you can use "Modbus Poll" input, as shown below
(CRC check generated automatically)



You can also use HyperTerminal serial input, as shown below
(Manually add CRC check)



1. Read AI analog input value:

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2))
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2))
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Modbus Address(PLC): 30001-30008

RS485 address : 0x01~0x3F

Function code: 0x04

Register address:0x0000-0x0007

Read number :0x0001-0x0008

For example, Read the value of AI analog input of channel 1-8 (channel 8 is 3.95MA, other channels are 0MA):

Send data(address 1): 01 04 00 00 00 08 F1 CC

Return data : 01 04 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 8B 14 DB

01 RS485 address, 04 function code, 10 length, 018B represents the value of channel 8, which is 395 in decimal, /100=3.95MA.

In addition, the AI analog input is also mapped to the 40000 interval register. The user can read the value of the AI analog input through the 03 function code.

Modbus Address(PLC): 40161-40168

RS485 address : 0x01~0x3F

Function code:0x03

Register address:0x00A0-0x00A7

Read number: 0x0008

For example, Read the value of AI analog input of channel 1-8 (channel 8 is 3.95MA, other channels are 0MA):

Send data(address 1): 01 03 00 A0 00 08 44 2E

Return data : 01 03 10 00 00 00 00 00 00 00 00 00 00 00 00 00
00 01 8B A5 AE

01 RS485 address, 03 function code, 10 length, 018B represents the value of channel 8, which is 395 in decimal, /100=3.95MA.

2. Set the AI scale value (correction value) :

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
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Modbus Address(PLC): 40193-40200

RS485 address : 0x01~0x3F

Function code:Write 0x06/0x10, Read 0x03

Register address:0x00C0-0x00C7

The voltage ratio can be corrected by this value when the voltage reading deviation is greater than 1%. The default value is 1000 (3E8).

For example 1: The actual voltage of channel 1 is 5.00V, but the read value is only 4.00V. The ratio deviation is $5/4=1.25$, and the correction voltage ratio is changed to 1250, which can correct the voltage.

Send frame: 01 06 00 C0 04 E2 0B 7F

Return frame: 01 06 00 C0 04 E2 0B 7F

The return frame is the same as the send frame. 07 means channel 1, 04

E2 means correction voltage ratio is 1250

For example 2: The actual voltage of channel 1 is 4.00V, but the read value is only 5.00V. The ratio deviation is $4/5=0.8$, and the correction voltage ratio is changed to 800, which can correct the voltage.

Send frame: **01 06 00 C1 03 20 D9 1E**

Return frame: **01 06 00 C1 03 20 D9 1E**

The return frame is the same as the send frame. 08 means channel 2, 03 20 means correction voltage ratio is 800

Special function Register

1.Set the 485 address(Slave ID)

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
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Modbus Address(PLC): 40254

RS485 address :0x01~0Xf8/0XFF

Function code:Write 0x06/0x10,Read 0x03

Register address:0x00FD(253)

Value: 2 bytes (values 1-248)

For example 1: Set the current device address to 0x02

Send data(address is 1): 01 06 00 FD 00 02 99 FB

Return data : 01 06 00 FD 00 02 99 FB

Send data(don't know the address): FF 06 00 FD 00 02 8C 25

Return data : FF 06 00 FD 00 02 8C 25

For example 2: Read device address(0X0001)

Send data : FF 03 00 FD 00 01 00 24

Return data : FF 03 02 00 01 50 50

Note: With this command, there can be only one module on the bus 485, More than one will go wrong!

2. Write baud rate

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16 (2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16 (2)
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Modbus Address(PLC): 40255

RS485 address :0x01~0x3F

Function code: Write 0x06/0x16; Read 0x03

Register address: 0x00FE (254)

Value: 2 bytes (values 0-7)

For example 1, Change the baud rate to 4800bps:

Send data(address 1): 01 06 00 FE 00 02 69 FB

Return data : 01 06 00 FE 00 02 69 FB

Baud rate corresponds to the number: 0:1200 1:2400 2:4800 3:9600

4:19200 5:38400 6:57600 7:115200 8: Factory reset

Note: 1 The baud rate will be updated only when the module is powered on again when this command is used!

2 When the number corresponding to the baud rate is 8, the factory settings can be restored

For example: 01 06 00 FE 00 08 E9 FC

For example 2 Read the current baud rate:

Send data(address 1): 01 03 00 FE 00 01 E5 FA

Return data : 01 03 02 00 03 F8 45

01 RS485 address, 03 Function, 02 length, F8 45 crc16, 03 means the current baud rate is 9600bps

Baud rate corresponds to the number: 0:1200 1:2400 2:4800 3:9600

4:19200 5: 38400 6:57600 7: 115200

3. Set Command(Date) Return Time

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2) (2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2) (2)
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Modbus Address(PLC): 40253

RS485 address :0x01~0x3F

Function code:Write 0x06/0x16;Read 0x03

Register address:0x00FC(252)

Value: 2 bytes (values 0-25)

For example, set the data return delay to 200ms

Send data(address 1):01 06 00 FC 00 05 89 F9

Return data :01 06 00 FC 00 05 89 F9

Return the delay time calculation formula: $X = 05 * 40 = 200MS$

Note: The maximum can be set to 1000MS. If it exceeds 1000MS, that is, the setting value is greater than 25, and the data return delay will be initialized.

That is: 01 06 00 FC 00 20 48 22 can make the data return delay to restore initialization 0

4. Set Parity

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2) (2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2) (2)
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Modbus Address(PLC): 40256

RS485 address :0x01~0x3F

Function code:Write 0x06/0x16;Read 0x03

Register address:0x00FF(255)

Value: 2 bytes (values 0-2)

For example, set the parity to Even parity

Send data(address 1):01 06 00 FF 00 01 78 3A

Return data :01 06 00 FF 00 01 78 3A

0 None Parity 1 Even Parity 2 Odd Parity

Note: 1. When using this command, the module is powered on again, and the check digit will be updated!

2. When the setting is greater than 2, the default value will be restored to 0 after powering on again, and there will be no verification.

5. Factory reset:

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
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Modbus Address(PLC): 40252

RS485 address : 0x01~0x3F

Function code:Write 0x06;

Register address:0x00FB(251)

Send data(address 1):FF 06 00 FB 00 00 ED E5

Return data :FF 06 00 FB 00 00 ED E5

For Command reset details, see: "AMIOA08 Factory Reset"

Link :https://1drv.ms/u/s!Av4PLxH_z8f1mBCFrTZWZSsyg4I?e=fgC5o

Hardware reset: Short the RES jumper of the board for 5 seconds, then power on again.